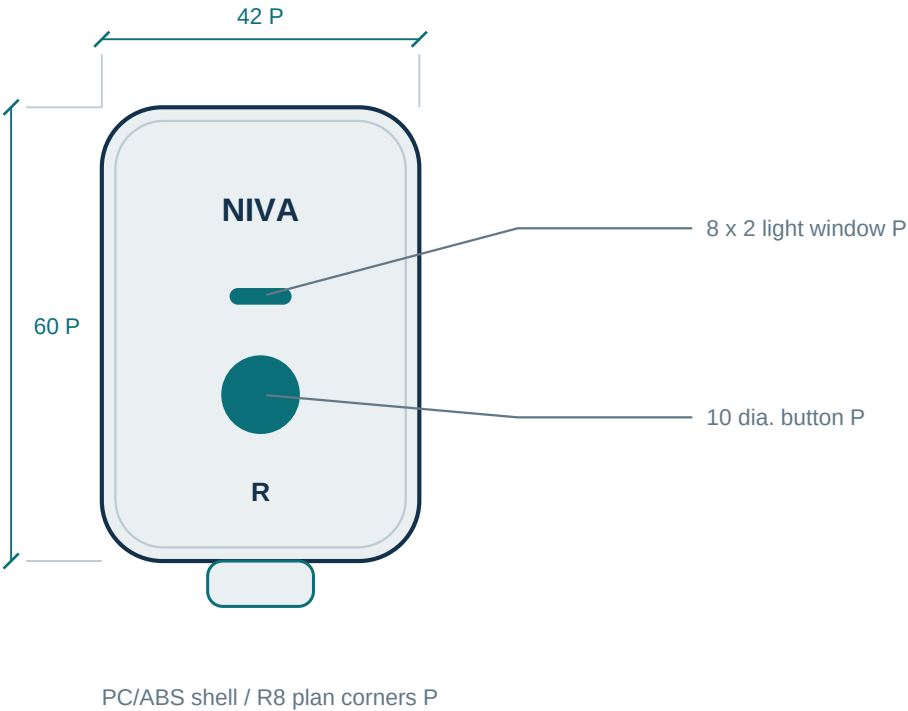
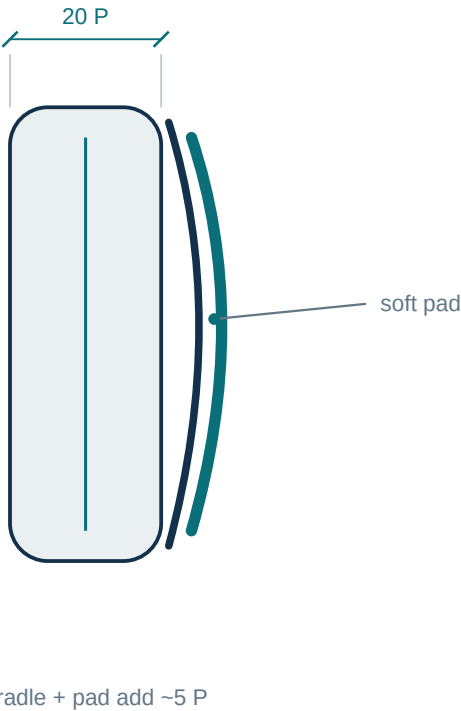


A calm, serviceable enclosure with a rear-removable battery and a stable lower-shin cradle.

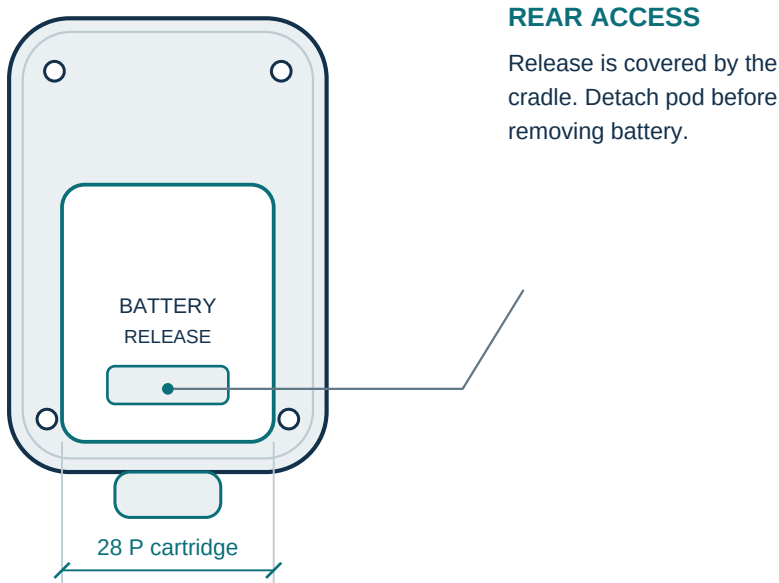
01 FRONT



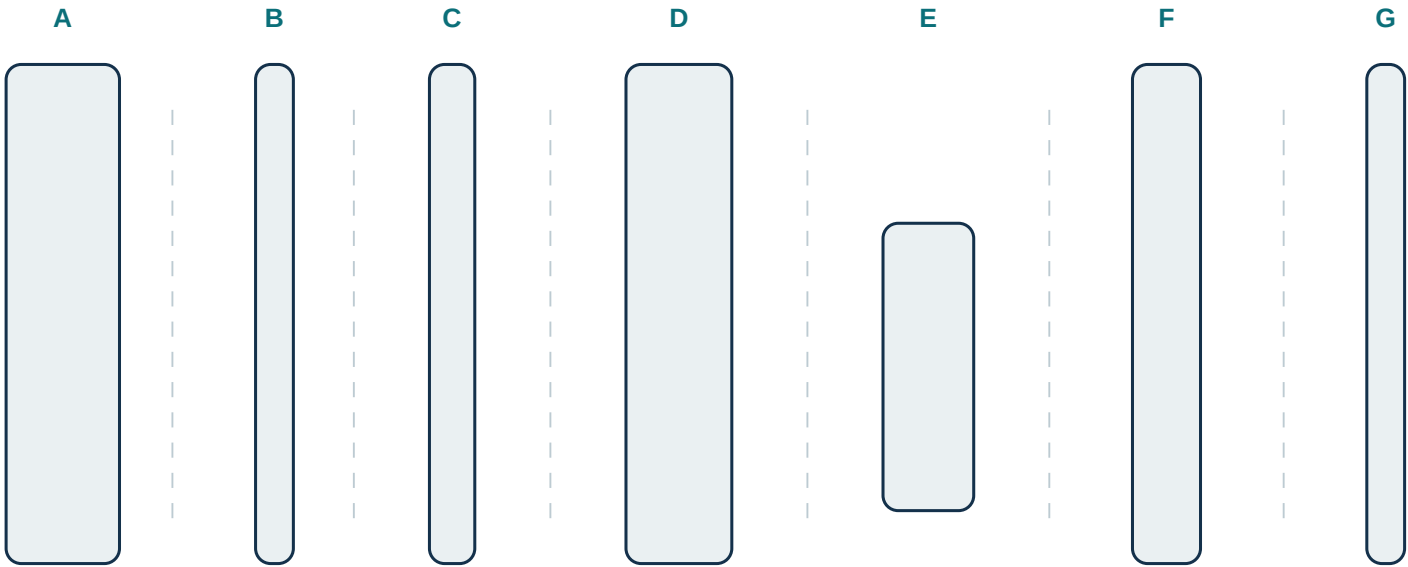
02 SIDE



03 REAR / CRADLE REMOVED



04 EXPLODED SECTION / SPACING EXAGGERATED



ASSEMBLY

- A Front shell + one-piece button membrane
- B Replaceable perimeter gasket
- C PCB fixed to rear shell; IMU cannot float
- D Rear shell with cartridge rails
- E Sealed, protected battery cartridge
- F Side-keyed cradle and 35-wide ladder strap
- G Replaceable smooth silicone contact pad

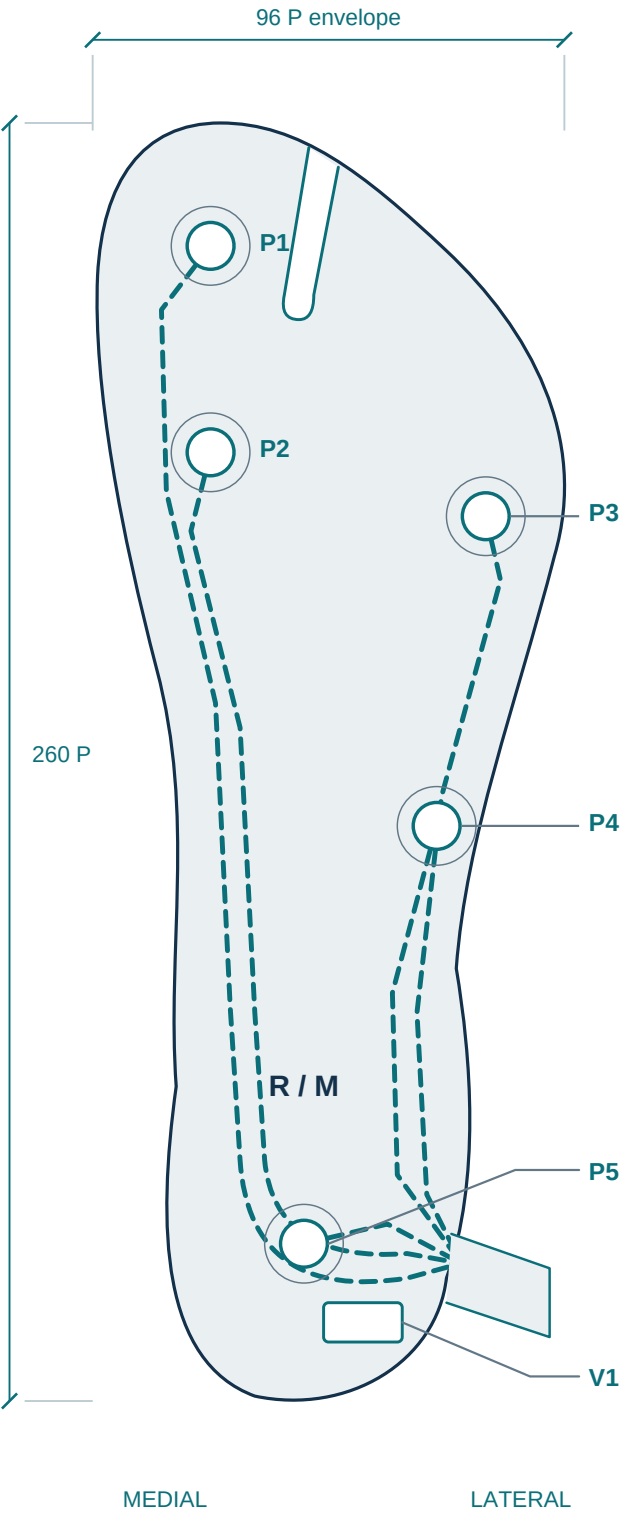
PACKAGING DECISION

60 x 42 x 20 P is a revised custom-PCB target, not a proven fit. The brief's 48 x 34 x 13 envelope is not carried forward. Breakout-board v0 needs a separate larger enclosure after parts are measured.

35 P strap width / soft radiused slots / flush rear service screws / no external charging port

Right medium insert shown. Left is mirrored. Sensor locations are design seeds, not measured anatomical fit.

01 SENSOR MAP



Dashed paths = routing intent only.
Sensor circles shown beneath top cover.

02 PLACEMENT / RIGHT FOOT

Datum: heel-most point, longitudinal axis parallel to drawing. X measured from medial envelope edge; Y measured forward from heel.

POINT	X / Y P	FUNCTION
P1	24 / 235	Big toe
P2	24 / 193	Medial forefoot
P3	80 / 180	Lateral forefoot
P4	70 / 117	Lateral midfoot
P5	43 / 32	Heel force

SENSOR GEOMETRY

A201 active diameter 9.53 V; thickness 0.203 V [S1].
Outer footprint, tails and recesses need supplier CAD. Do not substitute the active circle for a cutting outline.

TOE-POST ZONE

Soft slot 6 x 28 P; rounded termination. Keep traces outside a provisional 5-wide border. Validate toe-web fit across sizes. No field trimming in this revision.

HOLD H1 / STOCK TAIL REACH

The toe-to-heel route exceeds the standard A201 length (~190.5 overall). Stock tails cannot simply reach a common heel junction. Use bench sensors first; wearable routing requires supplier-approved long-tail/custom flex parts. No hard extension joints underfoot.

[S1] Tekscan A201: tekscan.com/products-solutions/force-sensors/a201 (checked 28 Sep 2026)

03 MATERIAL SECTION / SOFT STACK

0.4 P / Sealed TPU-faced cover
1.0 P / Continuous comfort foam
0.203 V / Sensor; recessed routing
1.0 P / Pocketed EVA / TPU base

Nominal ~2.6 + adhesive; target ~2.8 P.

V1: PVDF occupies a separate recessed heel zone beside P5. Its film, leads and cover must not create a thicker local stack.

NO HARD SPOTS

No pucks, pins, crimps, solder or boards in the plantar envelope. Use flexible film and continuous cushioning. Calibration must include the complete assembled stack.

SEAL + REPAIR

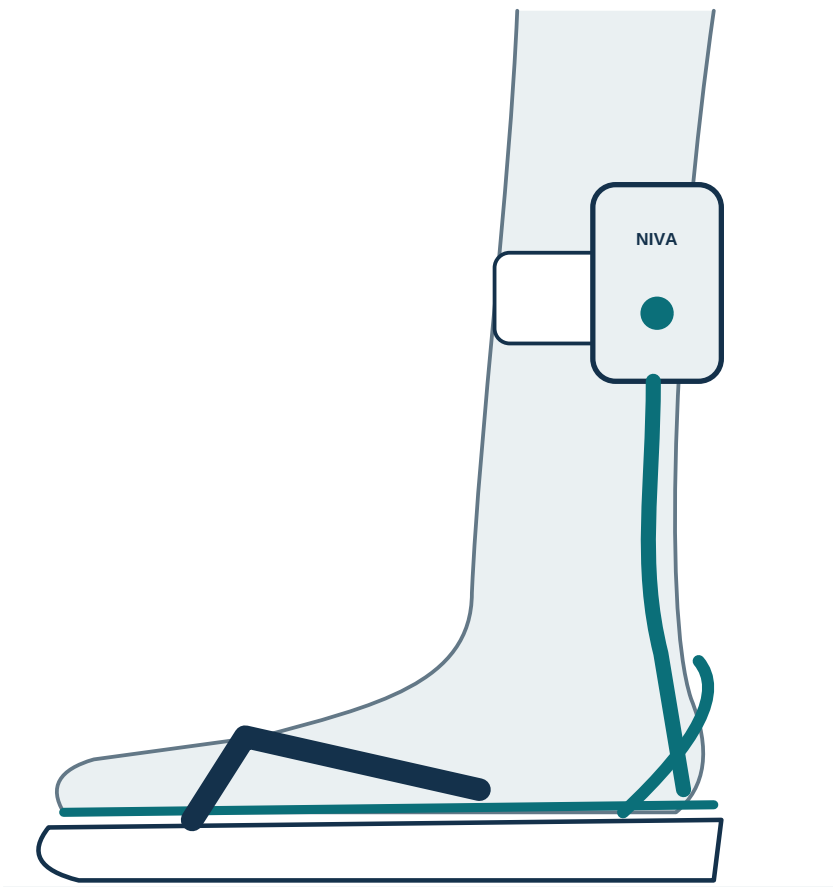
3 P perimeter seal allowance; process trials required. Replace the sealed insert and tail as one unit. Electronics, ID resistor and rigid junctions stay above the shoe collar.

BEFORE WEAR

Check full tail geometry, sensor overlap, seal peel, bending, sweat ingress, thickness and local pressure under repeated loading. The supplied 1:1 outline is a fit-study template only.

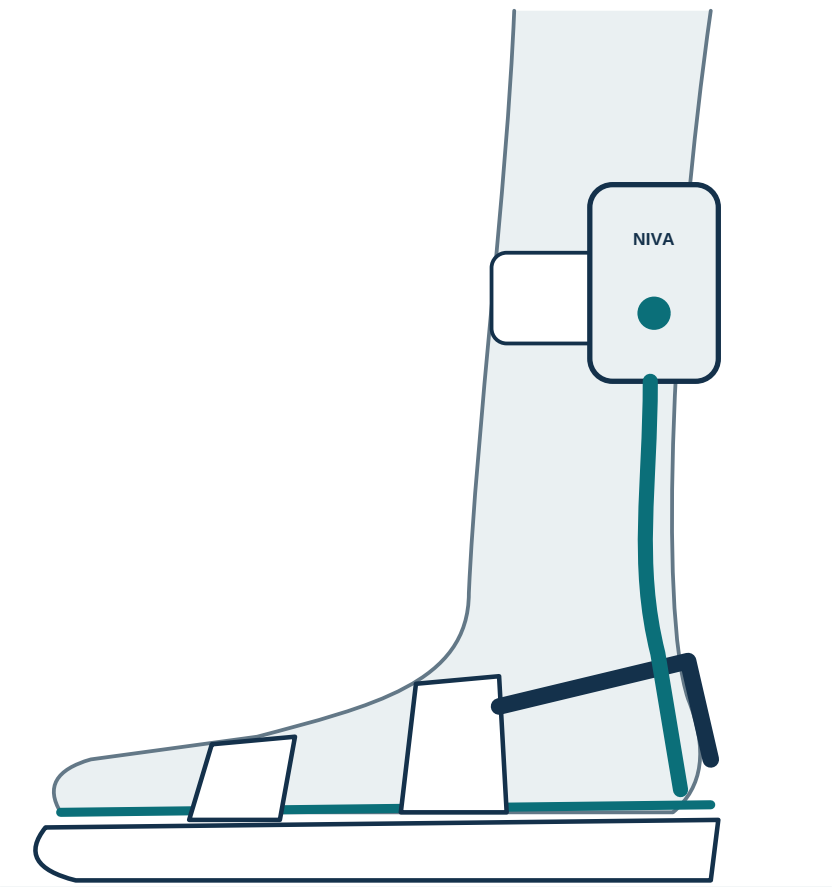
No rigid component sits in the shoe or beneath the foot. Retention changes; pod and sensing architecture remain common.

CHAPPAL



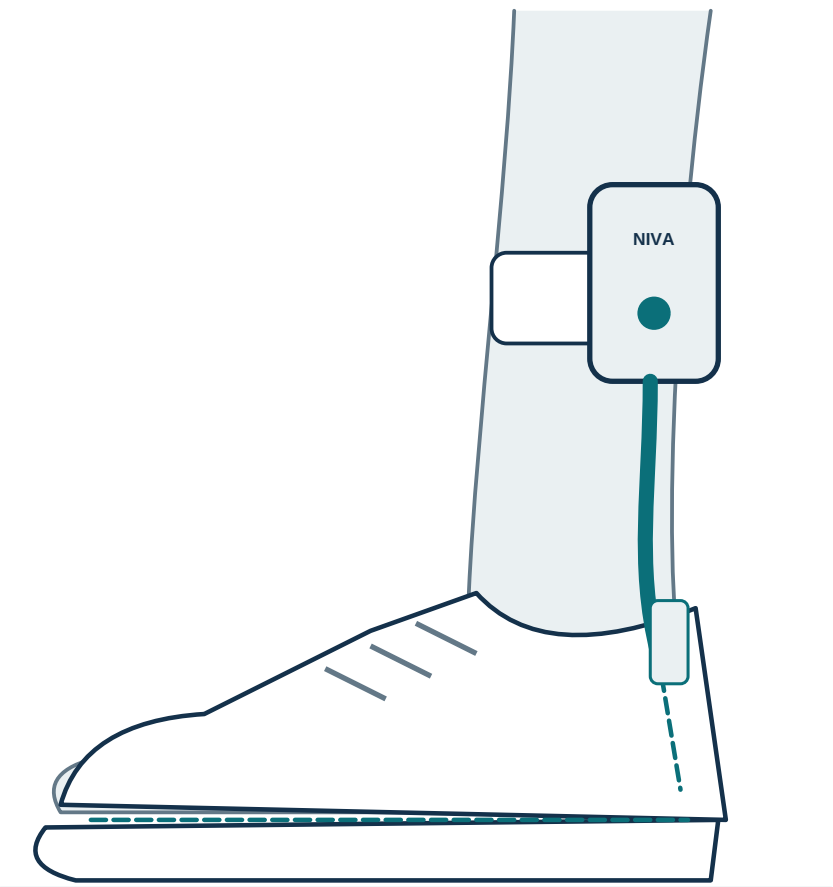
Toe-post slot clears the thong. Soft heel keeper locates the insert on the existing footbed. Validate migration.

SANDAL



Use existing straps. Route the soft tail clear of buckle and heel-strap pressure. Add keeper only if needed.

SHOE



Replace removable sockliner when possible. Sleeve protects the collar exit. Confirm volume and heel clearance.

GUMBOOTS + BAREFOOT

Use the supplied traction oversole if a boot shaft covers the pod or the shoe is too tight. Record the actual test footwear.

MOUNTING RULE

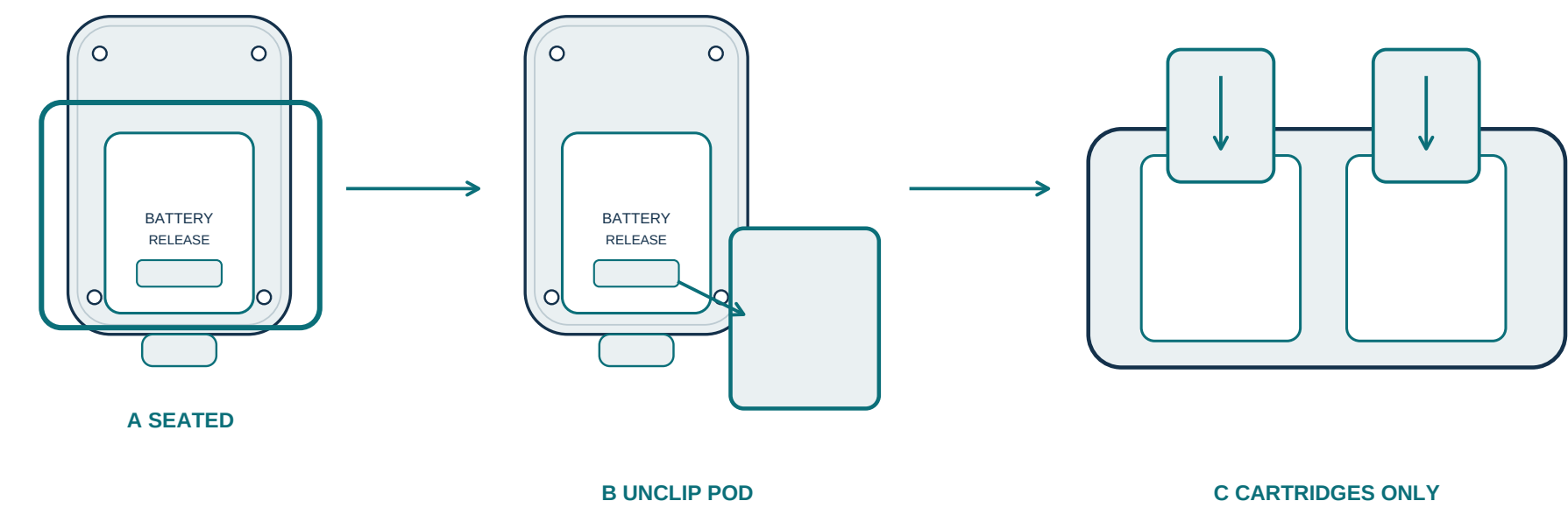
Cradle on anterolateral lower shin, clear of ankle bones. No fixed height fits everyone; confirm orientation and movement.

FIT CHECK

Seat insert; fasten pod; retain tail; verify anatomical side; check collar pressure; calibrate. Target <60 s, not yet proven.

Power is replenished only in a separate dock. The cartridge release is physically covered while the pod is seated.

01 REMOVE > RELEASE > DOCK



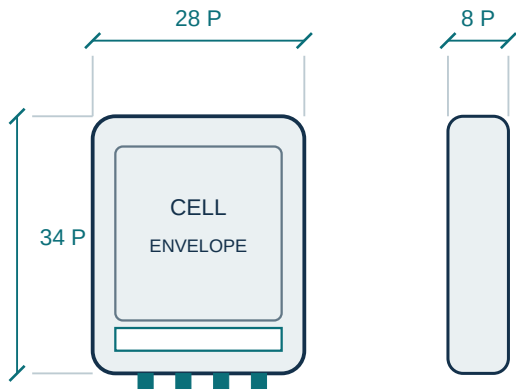
MECHANICAL EXCLUSION

Cradle masks the rear release. Battery cannot exit while seated. Pod body cannot fit the recessed charger wells. No charging path is provided through the sensor connector.

ELECTRICAL EXCLUSION

Protection resides in cartridge; charging resides in dock. Recessed contacts, polarity key and temperature sensing. Charger fault design still requires electrical review.

02 CARTRIDGE ENVELOPE



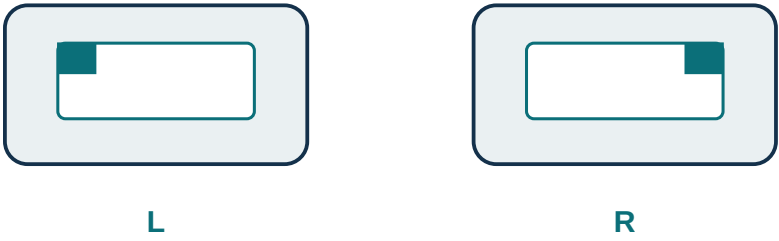
CONTACTS + CELL

Four protected contacts: B+, B-, NTC, ID. Cell envelope 22 x 23 x 5 P; capacity is unassigned until supplier selection. The brief's 300 mAh target is not a fit guarantee.

SERVICE DETAIL

Cartridge shell encloses the protected cell. No patient access to pouch or raw tabs. Rails, latch forces, seals and swelling allowance need supplier and mechanical review.

03 LEFT / RIGHT KEY INTENT



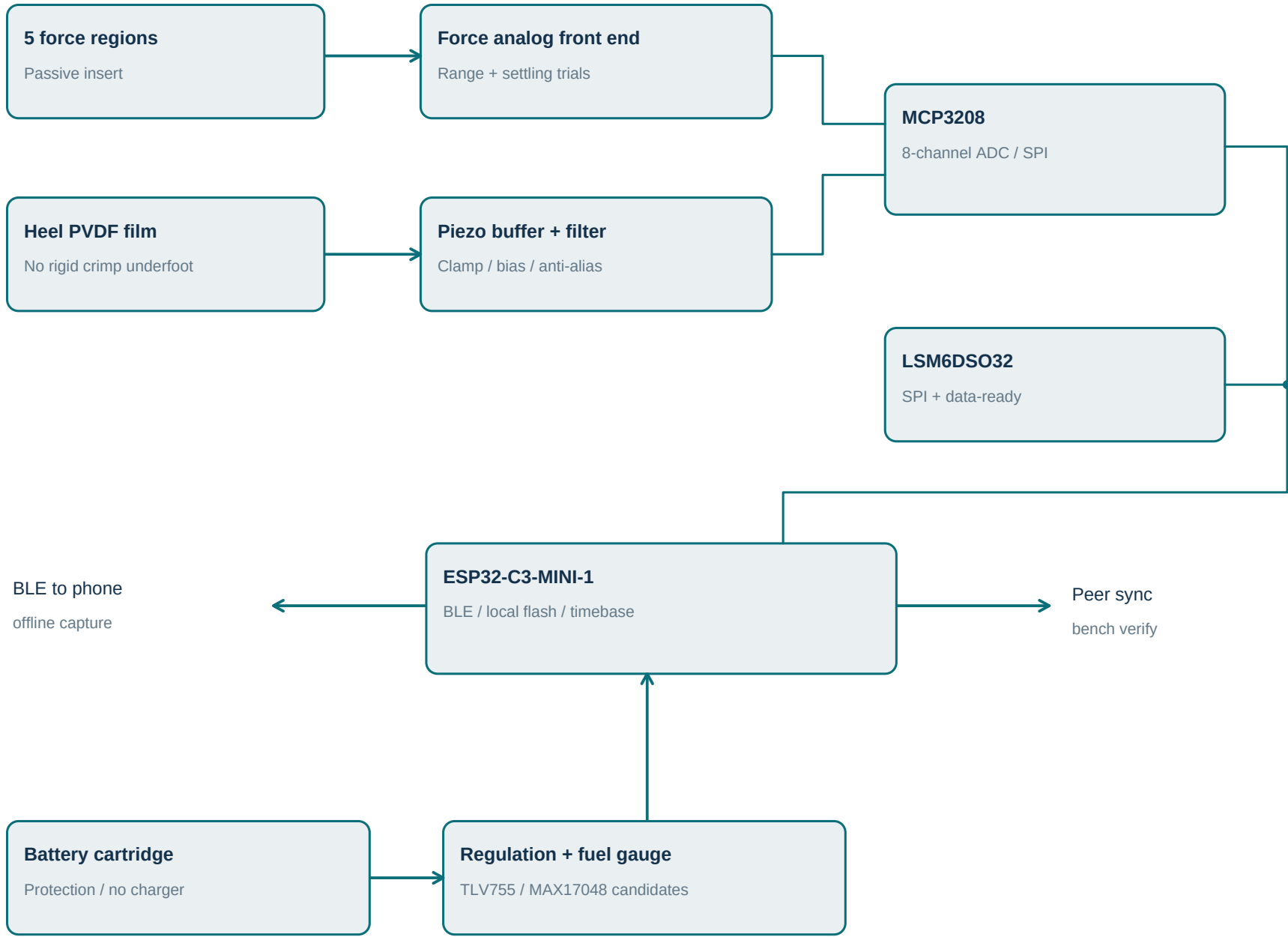
Different physical keys at the pod; readable L/R and tactile markers. Side/size ID resistor is in the above-collar connector assembly, paired to its insert.

RELEASE GATES

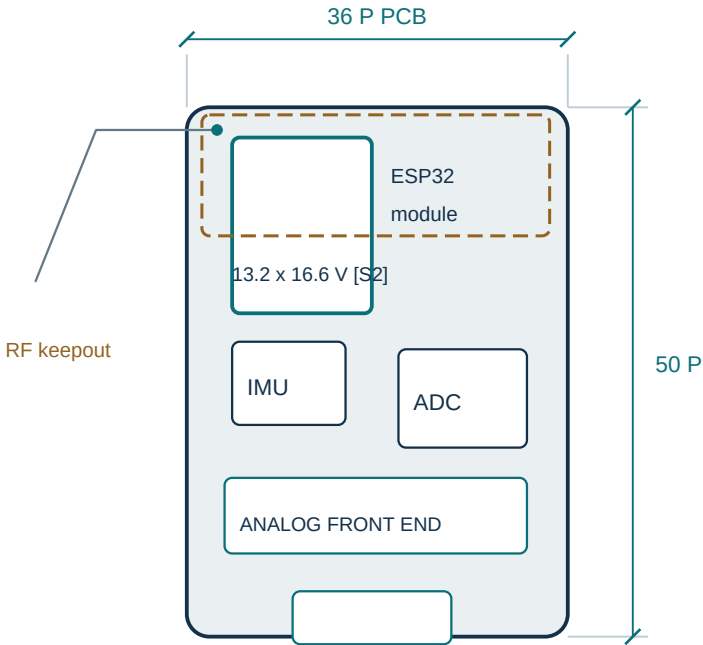
Confirm: no charge while seated; contact shorts; temperature fault; reversed insertion; latch wear; whole left kit fitted on right leg.

Architecture follows Claude's brief. This is a spatial and interface drawing, not a released schematic or PCB layout.

01 SIGNAL ARCHITECTURE



02 PCB / BATTERY ZONING



ZONE RULES

Antenna at top edge. No cell or metal over antenna region. Exact keepout per module integration guide. IMU rigidly fixed.

10-CONTACT STRIP / FUNCTIONAL MAP

1 VEXC; 2-6 FORCE1-5; 7 PZ+; 8 PZ-; 9 ID; 10 GND / shield drain. Final sensor drive topology may change the pinout. No power-input function.

ROUTING + SEAL

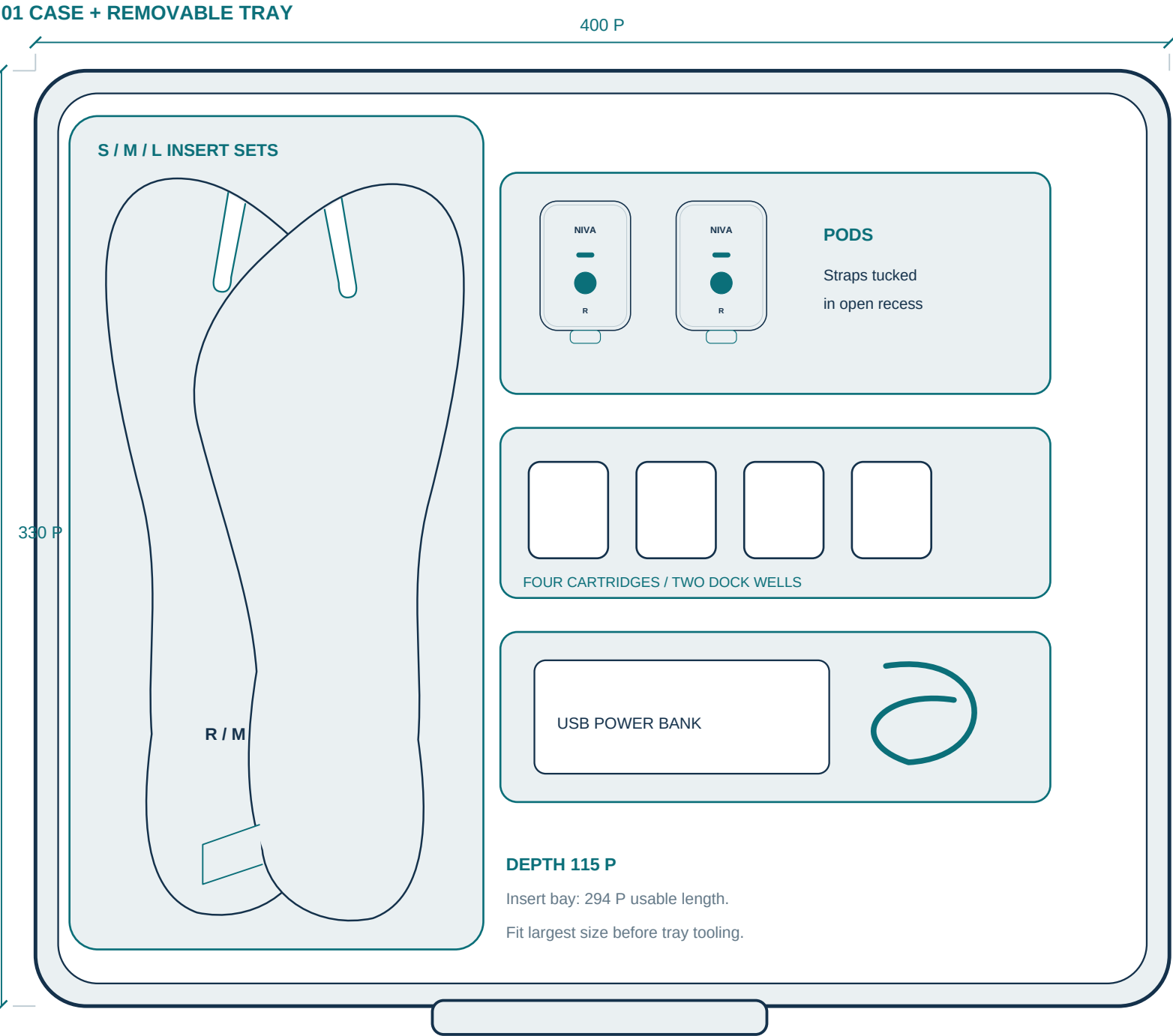
14-wide sleeve P; tail length selected to fit. Shield the piezo pair, retain slack in cradle, and keep every rigid junction above collar. Bend radius awaits cable selection.

ELECTRICAL REVIEW HOLDS / DO NOT COPY THE BRIEF INTO A PCB UNCHECKED

Verify sensor drive circuit, ADC source settling, boot-strapping pins with button held, RF coexistence, battery sag / regulator dropout, dock thermal-fault shutdown and sync accuracy. Manufacturer A201 performance uses an op-amp circuit [S1]; divider performance must be established separately.

[S2] Espressif ESP32-C3-MINI-1 datasheet v2.2: documentation.espressif.com/esp32-c3-mini-1_datasheet_en.pdf

A washable, modular field kit. The drawings define design intent while preserving the brief's bench-first development sequence.



TRAY + HYGIENE

Smooth removable polypropylene tray, open radiused recesses and no absorbent foam. Separate clean and used items during camp turnaround. Gasket and cleaning compatibility require tests.

CONTENTS

Two pods; size-matched insert pairs; two spare cartridges; dock; USB supply cable; power bank; replacement straps; heel keepers. Oversoles travel in a removable wipeable lid pouch.

- SERVICE PARTS**
- 01 Pod assembly
 - 02 Cradle + pad
 - 03 Strap
 - 04 Battery cartridge
 - 05 Insole + protected tail
 - 06 Heel keeper
 - 07 Traction oversole
 - 08 Dock + removable case tray

BASE KIT ONLY

Optional knee acoustic module is not included. No disease score or medial contact-force claim is embodied in the hardware drawings.

BUILD SEQUENCE / KEEP HIGH-RISK DECISIONS VISIBLE

01 BENCH

Sensor range, soft stack, tail noise and analog drive.

02 FIT STUDY

Print outline; measure feet; check footwear and migration.

03 WORKING V0

Measured breakout enclosure; log raw data and test faults.

04 ENGINEERED V1

Resolve H1 and component fit; then PCB, seals and tooling.